Nathan Brence
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Breakouts and Bricks: Prediction in the NBA

Which basketball efficiency metrics can most accurately predict future player performance?

Prologue

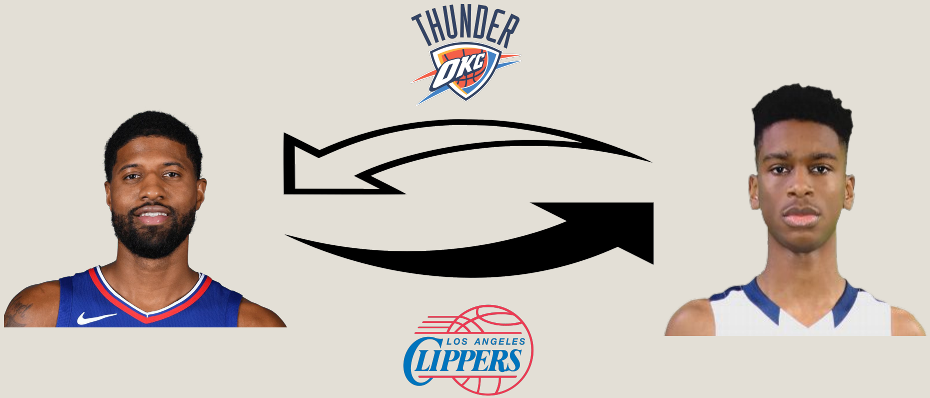
Motivation/Context, Goals, and Research Questions

Not All Fun and Games!

- Basketball = **EXTREMELY** expensive and exhaustive at the NBA level
  - Building a Winning Team: High salaries, complex trades, extensive scouting, player/front office negotiations, decisive draft picks, etc.
  - Small Mistakes = **BIG** Losses

- Predicting future performance: Also insanely difficult!
  - Intuition = **MYTH** (See Michael Jordan with the Charlotte Bobcats)

- What to do to increase efficiency?
  - Use the past to predict the future: Bring on the data analytics!



2025-2026 Player Salaries

RK	NAME	TEAM	SALARY
1	Stephen Curry, G	Golden State Warriors	$59,606,817
2	Joel Embiid, C	Philadelphia 76ers	$55,224,526
3	Nikola Jokic, C	Denver Nuggets	$55,224,526
4	Kevin Durant, F	Houston Rockets	$54,708,609
5	Anthony Davis, F	Dallas Mavericks	$54,126,450
6	Karl-Anthony Towns, C	New York Knicks	$54,126,450
7	Giannis Antetokounmpo, F	Milwaukee Bucks	$54,126,450
8	Luka Doncic, G	Los Angeles Lakers	$54,126,450
9	Jimmy Butler III, F	Golden State Warriors	$54,126,450
10	Jayson Tatum, F	Boston Celtics	$54,126,450
RK	NAME	TEAM	SALARY
11	Devin Booker, G	Phoenix Suns	$53,142,264
12	Jaylen Brown, G	Boston Celtics	$53,142,264
13	LeBron James, F	Los Angeles Lakers	$52,627,153
14	Paul George, F	Philadelphia 76ers	$51,666,090
15	Kawhi Leonard, F	LA Clippers	$50,000,000
16	Zach LaVine, G	Sacramento Kings	$47,499,660
17	Trae Young, G	Washington Wizards	$46,394,100
18	Lauri Markkanen, F	Utah Jazz	$46,394,100
19	Cade Cunningham, G	Detroit Pistons	$46,394,100
20	Jamal Murray, G	Denver Nuggets	$46,394,100

What Data Should We Use?

- General Intuition - Find data that corroborates player performance into a single measurable/comparative value
  - **Efficiency Metrics** attempt to do exactly this - Focus on them could be useful for prediction

Player Efficiency Rating (PER)

- "All-in-One" statistic; boils down player contributions to a single number
  - Average PER = 15; adjusted for each season
- Formula extremely long; contains box score, shooting %, PFs, etc.
- Cons
  - Heavily favors offense over defense
  - Unreliable with low MP players

Box Plus/Minus (BPM)

- Estimated points generated above average per 100 possessions played
  - Average = 0; adjusted for each season AND team
  - Estimates also weighed by position
- Cons
  - Struggles with defensive estimation
  - Only considers on-ball stats; off-ball contributions ignored

Value Over Replacement Player (VORP)

- Estimates player contribution "over replacement"
  - Subtracts BPM by replacement BPM (-2.0), multiplies by % of possessions and games played
- Cons
  - Highly correlated with BPM, (redundant?)
  - "Replacement Level" overinclusive; could be low for many reasons

Research Questions

1. **Are efficiency metrics (PER, BPM, VORP) significant estimators of future player performance in the NBA?**

2. **To what extent can the use of efficiency metrics as estimators of future player performance increase accuracy of prediction?**

3. **Which efficiency metric is best at predicting future player performance?**

Part One

Data and Variables of Interest

Collection Criteria

- Timespan: Regular seasons from 2010 to 2025
    - Why? Combination of high sample size and capturing modern era of basketball
    - Use latest regular season (2026) as test data
- For a season to be considered:
    1. Player must have participated in at least 45% of regular season games
    2. All season data must be readily available (no missing values)

Individual Information

- Personal Characteristics
  - Age (in years)
  - Games Played (as a proportion of team games)
  - Traded (Binary variable, 0 or 1)
  - Undrafted (Binary)
  - Draft position (as a proportion of players drafted before an individual)
- No weight or height categories; measurements are often stale
  - Mostly captured in player position

Note: All of these from previous season (can't use current season data to predict) except draft variables




Single-Player Statistics

- Position
  - Binary Variables for Guard and Forward
- Box Store Measurements (Per 36 Minutes)
  - Basic Five: Points, Assists, Rebounds, Steals, Blocks
  - Turnovers
- Advanced Statistics
  - True Shooting % (TS%): All-encompassing shooting efficiency
  - Usage Rate (USG%): Percentage of team plays used (possessions ended) by player
- Efficiency Metrics
  - PER, BPM, VORP

Team Statistics

- Season Averages
  - Offensive Rating (ORTG): Points generated per 100 possessions
  - Defensive Rating (DRTG): Points allowed per 100 possessions
- Win Percentage
- Strength of Schedule
- Team TS%

Note: If a player was traded, these values will be averaged based on each team's stats and the proportion of games played for each team

Differential Variables

We want to consider a player's last-season statistical values AND how they've changed. We've already captured the former in the variables we've reviewed, and movement can be found using previous season differentials

Previous Season Differential Formula (for any statistic S):

PSD = S(Last Season) – S(Two Seasons Ago)

Hence, differentials for every relevant variable (i.e. the differential gives us meaningful information, so none for Age, Traded, Draft Position, Undrafted, and Position variables) will be included for analysis

Dependent Variables

- Reminder: Use the past to predict the future
  - How to measure change in efficiency statistics? Use differentials
- Dependent Variables: **Current Season** differentials for selected efficiency metrics
  - Current season value previous season value
- What does this have to do with predicting season quality? Find out in the next section!

LAST_PER	LAST_BPM	LAST_VOR
10.5	-2.1	0
10.7	-2.9	-0.2
11.5	-1	0.1
9.7	-4.8	-0.1
12.9	-0.9	0.6
16	0.7	1.9
13.1	-3	-0.3
11.8	-2.2	0
12.5	-1	0.4
14.4	-0.8	0.5
11.8	-2.6	-0.2
17	1	1.4
14.5	-1	0.6
16.5	0.3	1.1
15.1	0.6	1.7
15.1	0	1
14.5	0.1	0.7
15.3	-0.6	0.9
19.5	2.1	2.1
16.8	1.3	1.9
11.5	-3.4	-0.2
9	-4.3	-0.3
11.6	-1.6	0.2
9.2	-4.2	-0.7
12.6	-1.7	0.1
9.4	-2	0
11.5	-1.5	0.2

PER_DIFF	BPM_DIFF	VORP_DIFF
-1.3	-0.1	-0.1
0.8	1.9	0.3
0.9	0.4	0.4
2.3	3.4	0.1
3.1	1.6	1.3
-2.9	-3.7	-2.2
-1.3	0.8	0.3
0.7	1.2	0.4
1.9	0.2	0.1
-2.6	-1.8	-0.7
-2.3	-1.1	-0.2
-2.5	-2	-0.8
2	1.3	0.5
-1.4	0.3	0.6
0	-0.6	-0.7
-0.6	0.1	-0.3
0.8	-0.7	0.2
4.2	2.7	1.2
-2.7	-0.8	-0.2
0.2	-0.1	-0.7
5.1	2.1	0.3
-1.2	-0.4	0.1
-2.4	-2.6	-0.9
3.4	2.5	0.8
-3.2	-0.3	-0.1
2.1	0.5	0.2
0.7	1.3	0.1

Data Summary

Between the Training and Test dataset:

- 4500+ individual seasons
- 1000+ unique players
  - All NBA Teams represented (plus some defunct groups)
  - Mean Age = 27.7 Years
  - Mean Games Played % = 84.5%

Mean (Standard Deviation) of Dependent Variables (differentials)

- PER: -0.187 (3.506)
- BPM: -0.051 (2.257)
- VORP: -0.00572 (0.924)

Are 4500 seasons enough for prediction? Let’s find out!

Part Two

Methodology and Tools

Outline of Procedure

1. Configuring Variables
   a. Generating all regressors
   b. Standardization
2. Run Regressions
   a. Four Types: OLS, Ridge, LASSO, Principal Component Analysis
   b. Get estimates from training data, then predict values on test data
3. Comparative Process
   a. Calculate out-of-sample error and R-Squared
   b. Compare DV estimates with actual values

$S = \{X_1, X_2, X_3 \ldots X_k\}$, where k = # of regressors

OLS
Ridge
LASSO
PCA

Brief Explanation of Regression

- Strategy for finding effect of regression X on target variable Y
- All data points can be modeled by pictured regression equation
  - We're interested in finding estimates for $B_1, B_2, \ldots B_n$; the population effects of $X_1, X_2, \ldots X_n$ on Y
- How do we find these estimates? Many ways to do so!

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n + \varepsilon$$

Y = Dependent Variable

$B_1, B_2, \ldots B_n$: Size of effect of $X_1, X_2, \ldots X_n$ on Y

$X_1, X_2, \ldots X_n$: Regressors

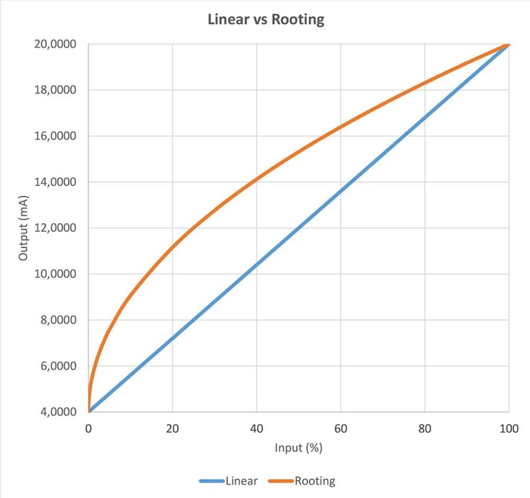
ε = Error Term, purely random influences on Y

1. Configuring Variables

Use data to construct three categories of regressors

1. Raw Predictors (X_1)
   - Simple, just the variables as collected
2. Square Regressors (X_1^2)
   - Raw predictors squared; helps capture nonlinear relationships
3. Interactions (X_1*X_2)
   - One variable multiplied by another
   - Ex.: What additional effect does high USG% have on PTS per 36 minutes?

After eliminating perfectly multicollinear variables, we're left with more than 600 regressors!



1. Configuring Variables (cont.)

- Each of our regressors follows a separate distribution - Counterintuitive for prediction!
- We solve this with standardization: All regressors are modelled to a standard normal distribution
    - Mean = 0, Standard Deviation = 1

- For Dependent Variables: Standardize, but keep unstandardized variants for additional regressions
    - Why? Compare prediction error in units of specific efficiency metrics AND in standardized levels across metrics

$$z = \frac{x - \mu}{\sigma}$$

Standardization Formula

Z = Standardized datapoint
X = Unstandardized datapoint
μ = Variable Mean
σ = Variable Standard Deviation

2. Run Regressions

- We’ll run twenty-four separate regressions
  - Four methods for our six dependent variables (standardized and unstandardized efficiency metric differentials)
- Use estimates for each B in our out-of-sample test data (2025-26 NBA season) to get predicted dependent variables

Note: All regressions were performed using Stata

RAW DATA

Configure Variables for Regression

Obtain In-Sample Coefficients

OOS PREDICTIONS

Regression Structure

Each regression will include the same non-efficiency metric regressors, but will only include the efficiency metric variables that match the dependent variable

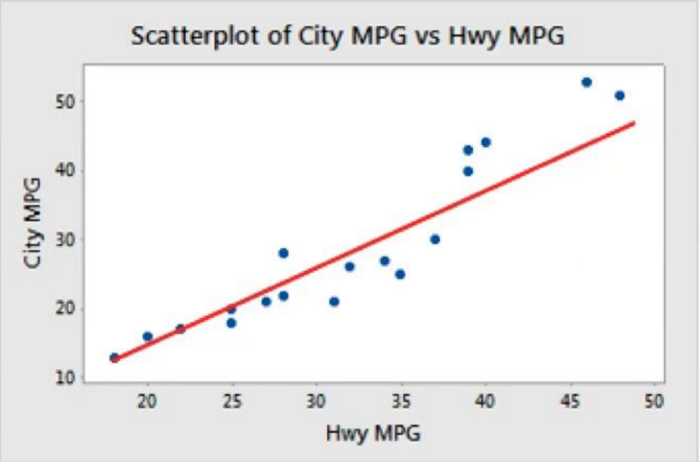
So regressions with PER DVs will only include PER regressors in addition to the non-efficiency metric variables. Each regression will include exactly **519** regressors

Why? Isolate predictive capacity of each efficiency metric

Types of DV	PER DVs	BPM DVs	VORP DVs
Included Regressors	Non-efficiency metric regressors + PER raw predictors, square variables, and interactions	Non-efficiency metric regressors + BPM raw predictors, square variables, and interactions	Non-efficiency metric regressors + VORP raw predictors, square variables, and interactions

Regression Methods: Ordinary Least Squares (OLS)

- Choose B to limit Sum of Square Residuals (SSR)
  - In other words, minimize sum of square distances of data points from regression line
- Great for predicting causal effects, but not the best for prediction
  - We'll use OLS as a baseline for more efficient predictive models



$$SSR = \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2$$

$\hat{y}$ = predicted dependent variable

$\bar{y}$ = mean of dependent variable

Ridge Regression

- Similar to OLS, but minimizes a “penalized” sum of square residuals
  - Added “penalty term” shrinks large estimations of B (common with OLS) to zero given the parameter lambda
- Idea: Reducing larges estimates = reducing variance, which then leads to lower prediction error
- How to choose lambda?
  - M-fold Cross Validation: Find lambda that produces lowest mean squared error

$$S^{\text{Ridge}} = S^{\text{OLS}} + \lambda_{\text{ridge}} \sum_{j=1}^{p} \beta_j^2$$

S^{Ridge} = Ridge SSR

S^{OLS} = OLS SSR

λ^{Ridge} = Penalty Parameter

LASSO Regression

- Same structure as Ridge: Included penalty parameter reduces large estimates of OLS
  - Note the difference in the parameter: Sum of absolute values of estimates
  - Implies that many estimates will be set to ZERO
  - Is this bad? Not necessarily!
- Determine lambda through MFCV

$$S^{\text{Lasso}} = S^{\text{OLS}} + \lambda_{\text{lasso}} \sum_{j=1}^{p} |\beta_j|$$

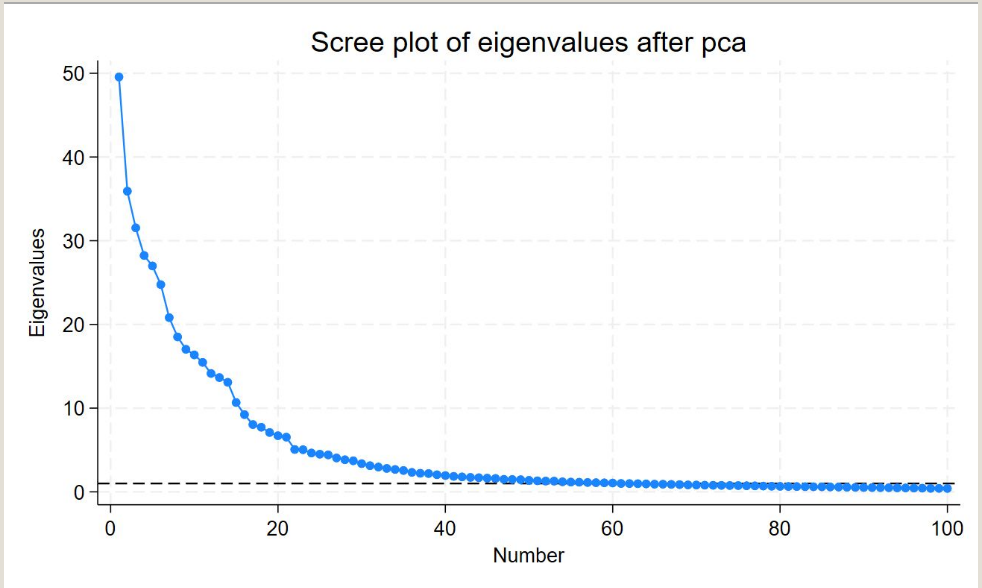
S^{LASSO} = Lasso SSR

S^{OLS} = OLS SSR

λ^{LASSO} = Penalty Parameter

Principal Components (PCA)

- Capture variation in linear combinations of data, which are then used as regressors
  - Reduce number of regressors by using only certain portion of IV variation
  - For this project: Use enough PCs to capture 95% of variations
    - PER: 75 PCs
    - BPM: 80 PCs
    - VORP: 95 PCs



Variation captured given a number of principal components

3. Comparative Process

- Take in-sample estimates to out-of-sample data
  - Standardize OOS variables with IS means/SDs!
- Generate OOS predictions for dependent variables, calculate RMSE and R-Squared
  - RMSE = Root Mean Squared Error: Average prediction deviation from true value
  - R-Squared: Proportion of variation explained by regression

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(\hat{y}_i - y_i\right)^2}$$

$$R^2 = 1 - \frac{SSR}{TSS}$$

TSS = Total Sum of Squares; variance of the dependent variable

3. Comparative Process (cont.)

To categorize season quality, we'll take the mean and standard deviation of our OOS dependent variables and define each classification as such:

Breakout: >+1.5SD from Mean
Improvement: Between +0.75SD from +1.5SD from Mean
Plateau: Between -0.75SD from +0.75SD from Mean
Worsening: Between -0.75SD from -1.5SD from Mean
Notable Decline: >-1.5SD from Mean

Using these guidelines, both the actual and predicted OOS values will be classified and then compared, with the percentage of correct predictions over the "Naive Model" being reported

Naive Model: Every prediction is "Plateau"

Part Three

Results

Bold = Best of DV
Italic = Best of all DVs

Out-of-Sample Root Mean Squared Error

	OLS	Ridge	LASSO	PCA
PER Standardized (Unstandardized)	0.749 (2.625)	***0.714*** (2.494)	0.716 (2.497)	0.725 **(2.254)**
BPM Standardized (Unstandardized)	0.854 (1.936)	**0.839** **(1.895)**	0.844 (1.899)	0.856 (1.955)
VORP Standardized (Unstandardized)	0.901 (0.833)	0.859 (0.793)	**0.848** **(0.781)**	0.866 (0.802)

- Shrinkage Estimators have lowest RMSE - not all that surprising
  - Unstandardized RMSE is a bit unsettling given the scale of each metric; even lowest BPM error is difference between average and replacement level

Bold = Best within DV
Italic = Best of all DVs

Out-of-Sample R-Squared

	OLS	Ridge	LASSO	PCA
PER	0.0393	**0.1271**	0.1242	0.1007
BPM	0.1986	***0.2268***	0.2186	0.1959
VORP	0.1094	0.1891	**0.2102**	0.1775

- PER has the lowest overall R-Squared (so much for being an all-in-one statistic?)
- BPM and VORP extremely similar (expected, since the former is an input of the latter
  - Again, shrinkage estimators display highest explanation of variation
  - Maps perfectly to RMSE leaders; R-squared and RMSE work hand-in-hand

Bold = Best within DV
Italic = Best of all DVs

Correct Prediction % Above Naive Model

	OLS	Ridge	LASSO	PCA
PER	-6.53%	-8.25%	-6.82%	**-5.50%**
BPM	-7.90%	-6.19%	**-5.50%**	**-5.50%**
VORP	-6.53%	-2.06%	-1.72%	***0%***

- Not one model predicts better than the naive classifier: Bad News!
- Despite not leading in RMSE or R-Squared, PCA takes a lead in prediction accuracy
    - Likely not significant difference, however
    - In addition: Most accurate PCA model predicted **ZERO** breakout seasons correctly

Part Four

Conclusions and Further Research

Answering Our Research Questions

1. **Are efficiency metrics (PER, BPM, VORP) significant estimators of future player performance in the NBA? No!**

2. **To what extent can the use of efficiency metrics as estimators of future player performance increase accuracy of prediction? Minimal/No Extent**

3. **Which efficiency metric is best at predicting future player performance? Technically: VORP; Practically: NONE**

Conclusion

PER, BPM, and VORP are not accurate predictors of <u>ANYTHING</u>

Design Improvements and Considerations

- More Regressors!
  - Injury Information
  - Post-Season Awards
  - Playoff Binary
- More Efficiency Metrics!
  - xFG%+, RAPM
- More Regressions!
  - If we're interested in classification, Random Forest Regression may be usable
- Classification System
  - Bounds are a bit arbitrary, but reasonable
  - Classify seasons based on identical, independent metric

Possible Further Research

- Include Playoff Information
  - Combine Regular/Postseason PER/BPM/VORP, weight accordingly
- Creation of New Metrics
  - Focus on most relevant predictive factors and use them as inputs for new metrics
- Implement Metrics from other sports
  - Could WAR (WIns above Replacement), which is common in baseball, be used in an NBA context?

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Thank You For Listening!

(and Go Spurs!)